

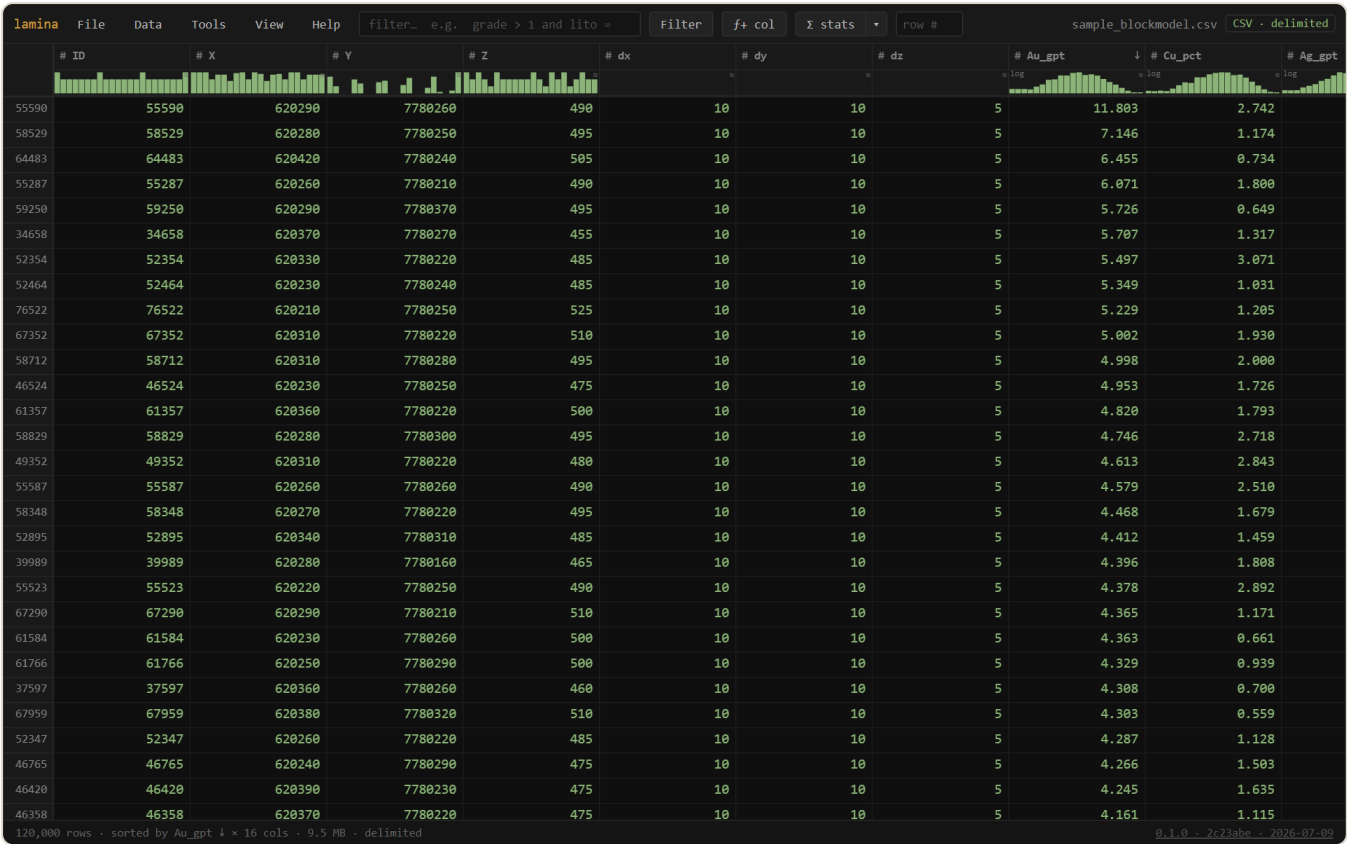
lamina

Open any file, however large — scroll, filter, sort, derive, summarize, and quality-check it. Windowed, read-only, offline.

 **Sealed** — no network, nothing leaves your machine · single self-contained HTML file · [download lamina.html](#)
· [open the app](#) ↗

[User documentation](#) · [Auditable](#) / [Geoscientific Chaos Union](#) · gentropic.org/lamina

lamina never loads the whole file, so size isn't the problem: a multi-gigabyte CSV or Datamine `.dm` scrolls in about a second, filters and sorts in place, and reports grade–tonnage — all without an import step, a server, or a byte leaving your machine.



A block model open in lamina. Every column header carries a live distribution glyph — histogram for numbers, top-values bar for categories — and the grid scrolls the file windowed, never resident.

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Reference

1 What lamina is

lamina is a single self-contained HTML file — a **windowed data viewer** built for the files resource geology actually produces. It reads over a coarse block index, decoding only what's on screen (plus what a scan is currently sweeping), so the first look is near-instant at any size and memory stays flat.

It reads:

- **Delimited tables** — CSV / TSV / TXT / DAT, any size; GSLIB / Geo-EAS and whitespace dumps detected, `#` comment preambles skipped
- **Datamine** `.dm` — opened directly at any size, decoded on the fly (no conversion), with the same filter / sort / stats as CSV; filters read only the columns they need (strided), which is how a 17-column file filters 30× faster than a naïve decode
- **Archives** — zip / tar / gz / zst / bz2: peek inside and window huge compressed entries *without unpacking*
- **Text** → lines · **binary** → hex — nothing refuses to open

Everything is **read-only**: lamina never modifies your file. Derived things — calculated columns, filters, formats — live in the view (and can be saved as a lens), while Export writes a new file.

Tip — No file handy? The empty screen offers ▶ *open a sample block model* — 120,000 rows generated locally, nothing downloaded.

2 Getting started

File → **Open** (`Ctrl+O`) or drag a file anywhere onto the window. Recent files are remembered (toggleable) and reopen with one click.

If a file reads wrong

Click the **kind badge** (top right) or **Data** → **Interpretation...** to force the delimiter, header on/off, rows to skip, comment prefix, decimal point vs comma, or the **character encoding** (UTF-8 · Windows-1252 · Latin-1 · UTF-16). A `⚠ encoding?` hint appears in the footer when bytes don't decode cleanly as UTF-8 — mojibake never passes silently.

3 The grid

The grid is windowed: scrolling a 1.8-million-row file reads just the rows in view. **Click a header** to sort (ascending → descending → off; sorts can stack across columns), the **row #** box jumps, and `Ctrl+F` opens **find** (case / whole-cell / regex / per-column scope / match count). Selected cells copy as TSV with `Ctrl+C`.

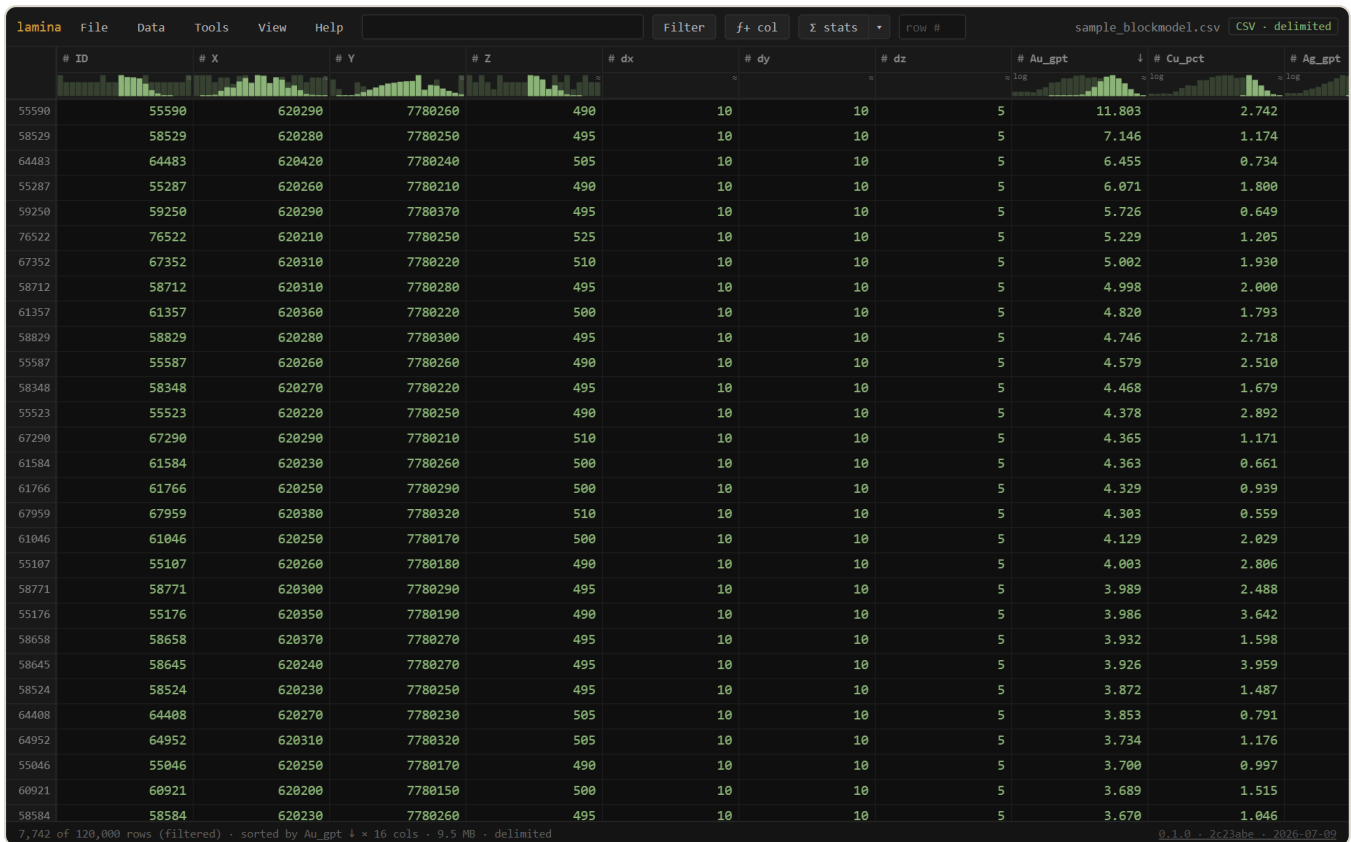
Two dockable panels (View menu): the **Columns** panel — search, show/hide, drag-reorder, pin-freeze, and per-column settings — and the **Record inspector** — one row's fields as a list that follows the selection; `↑`/`↓` step rows and `⌘` **pin** holds one for a side-by-side compare.

Right-click a numeric header → **Color scale** heatmaps the cells (8 palettes, linear/log, clip outliers, reverse) — patterns in a million rows read at a glance. **Number format** and **treat as text / number** live in the same menu.

4 Column distributions

Each column header carries a live **distribution glyph**: a histogram for numbers (with a null-rate bar; $\approx$ means sampled, `log` means auto log-scaled for skewed grades), a coloured top-values bar for categories. They're instruments, not decoration:

- **hover** — read the value / category under the cursor
- **click** — filter to that bin or category
- **drag** — brush a range → a filter (auto-applies when cheap, stages for `Enter` on huge files — Data → Gutter brush)
- **double-click** — the column's full Statistics



Filtering: an expression in the box — or a drag across a header glyph — and the grid shows only the matches, windowed as ever. The footer counts them.

5 Filter

The filter box speaks a **SQL-WHERE**-style language: `grade > 1 and lito = "OXIDE"`. `Enter` applies, `Esc` clears. Highlights: `between ... and ...`, `in (...)`, `contains` / `like` / `matches` for text, `is blank` / `is filled`, functions (`round sqrt log if(c,a,b) bin coalesce isnum ...`), and parentheses. Bare words are *columns* (so `fe > cu` compares two columns); quoted words are text. Blanks behave sanely — no SQL `NULL` traps. Autocomplete offers columns, functions, and data values.

On a `.dm` file the filter reads *only the columns the expression uses* — the strided fast path — so a selective filter over a huge model finishes in seconds.

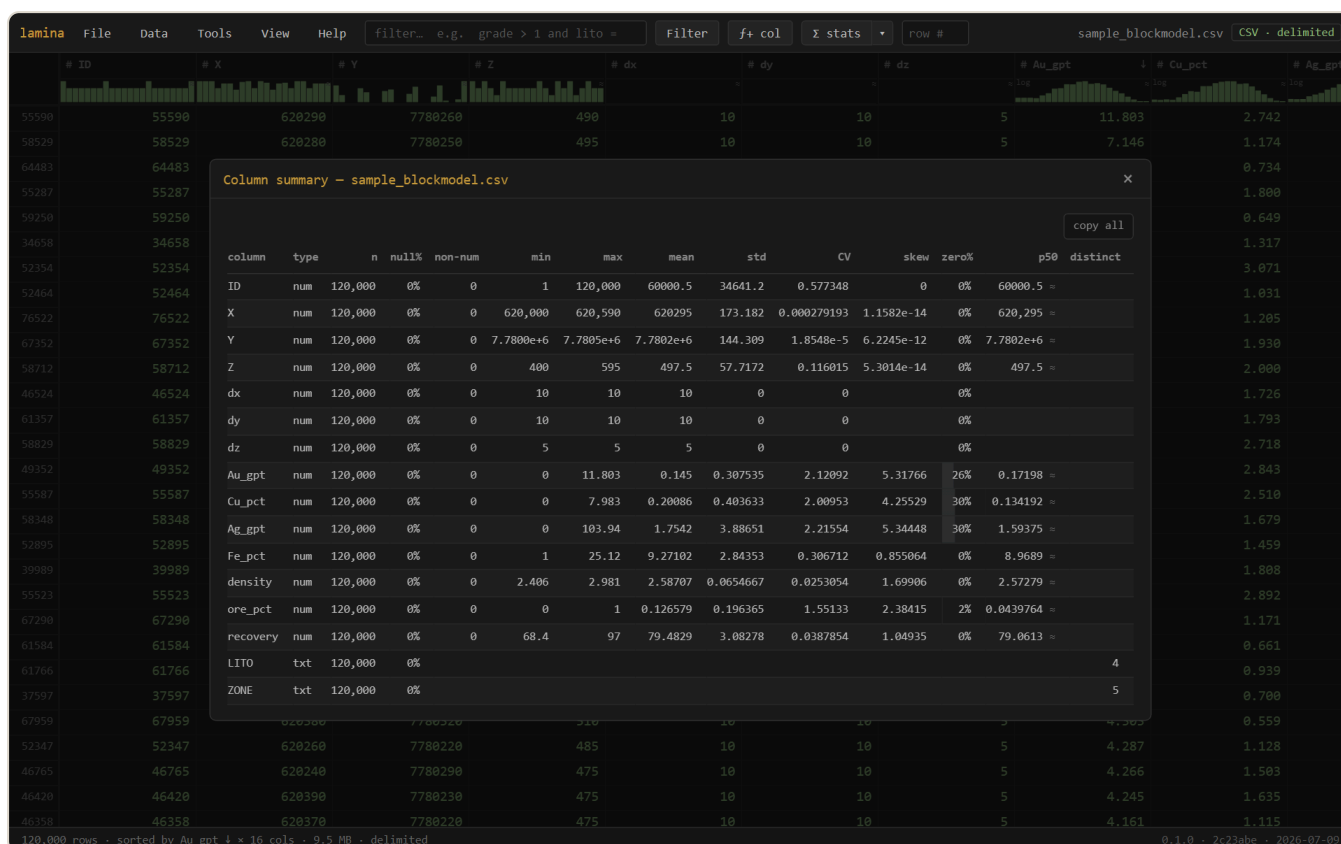
6 Calculated columns

The **f+ col** button (or a header's right-click, or **Data** → **Calculated columns...**) adds a derived column from a formula in the same language as the filter: `grade * density`, `if(au > 1, "ore", "waste")`, `bin(rl, 10)` for elevation bands. Computed on the fly, never written; filter, sort, stat, group, and export it like any column.

7 Statistics

One column: double-click its glyph (or right-click → Statistics) — histogram with a log-x toggle, quantiles, count / nulls / non-numeric (with examples), min / max / mean / std / **CV** / **skew** / zeros, and **exclude zeros / negatives** re-stats for grade-only views. Copies as TSV.

Every column: **Σ stats** on the toolbar sweeps the whole table once and opens the **Column summary** — a sortable row per column (n · null% · non-num · min · max · mean · std · CV · skew · zero% · p50). Sort by **null%** or **non-num** to triage a new file in seconds; click a row for detail, click a name to jump to the column in the grid.



Σ stats — the whole file, one pass, one sortable table. The fastest way to meet a file you've never seen.

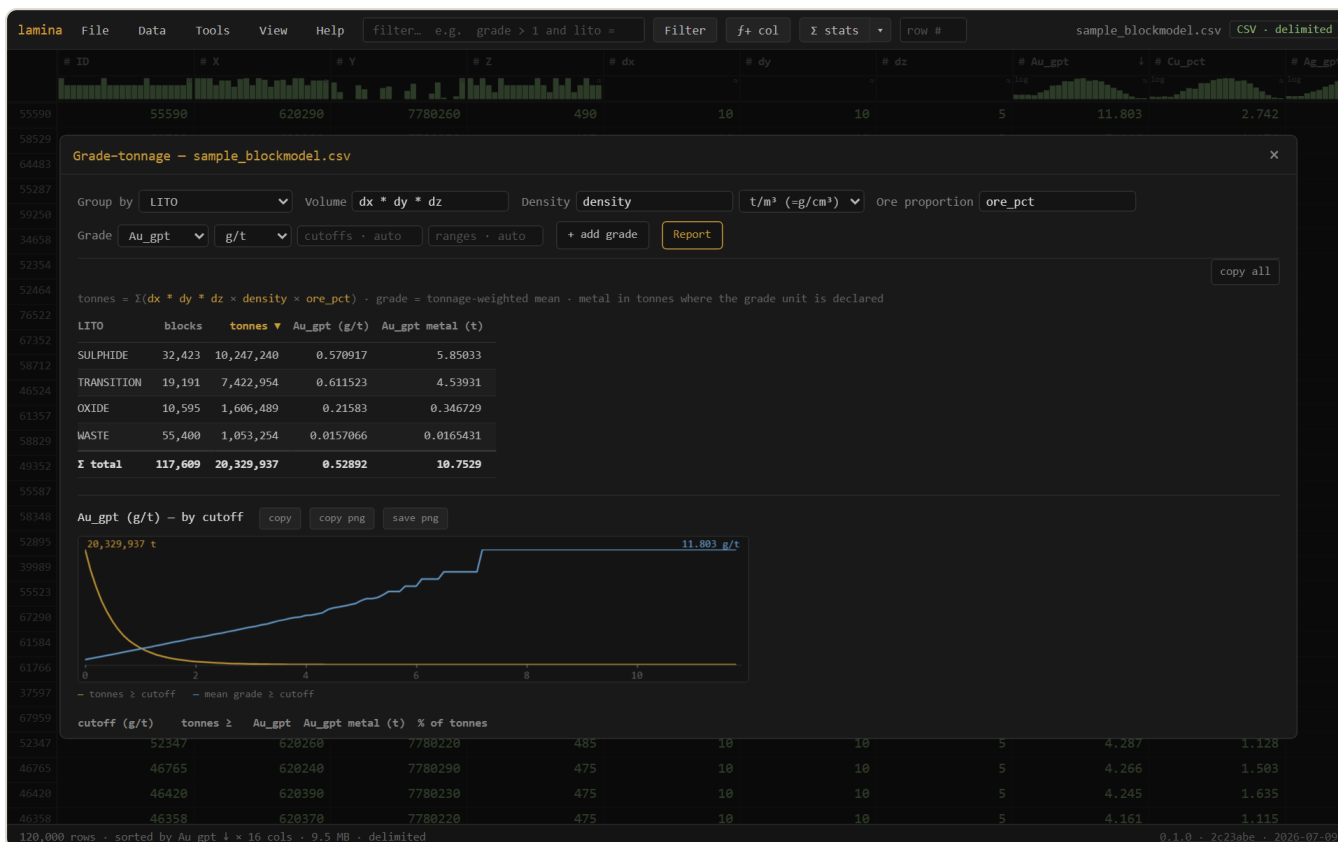
8 Group by

Tools → **Group by...** aggregates value columns per group, with an optional **weight** column — the weighted mean shows next to the plain one, so an unweighted-vs-length-weighted grade difference is visible instead of hidden. Click a group to filter the grid to it; copy as TSV. Numeric bands: add `bin(rl, 10)` as a calculated column and group by it.

9 Grade-tonnage

Tools → **Grade-tonnage...** is the resource view. **Volume · density · ore proportion** are expressions (a column, a constant, `dx * dy * dz` — auto-detected where obvious; blank = 1), with a **declared density unit** so a kg/m³ column doesn't inflate tonnage a thousandfold. The report gives **tonnes · tonnage-weighted grade · contained metal** per domain (+ Σ total), and every grade field gets its **cutoff curve** — tonnes $\geq$ cutoff falling, mean grade rising — with a table at automatic round cutoffs *or your own* (`0.5`, `1`, `2` — computed exactly, not snapped).

- **Declared grade units** (% , g/t , ppm , oz/t) label the report and make *metal come out in tonnes*.
- **Manual plot ranges** per grade — `x:0..2 t:5e6 g:3` pins the cutoff window and the y-scales, so this month's curve overlays last month's at the same scale.
- Each curve offers **copy** (TSV) · **copy png** · **save png**.
- Respects the current filter, and computes windowed — a multi-GB model reports without loading.



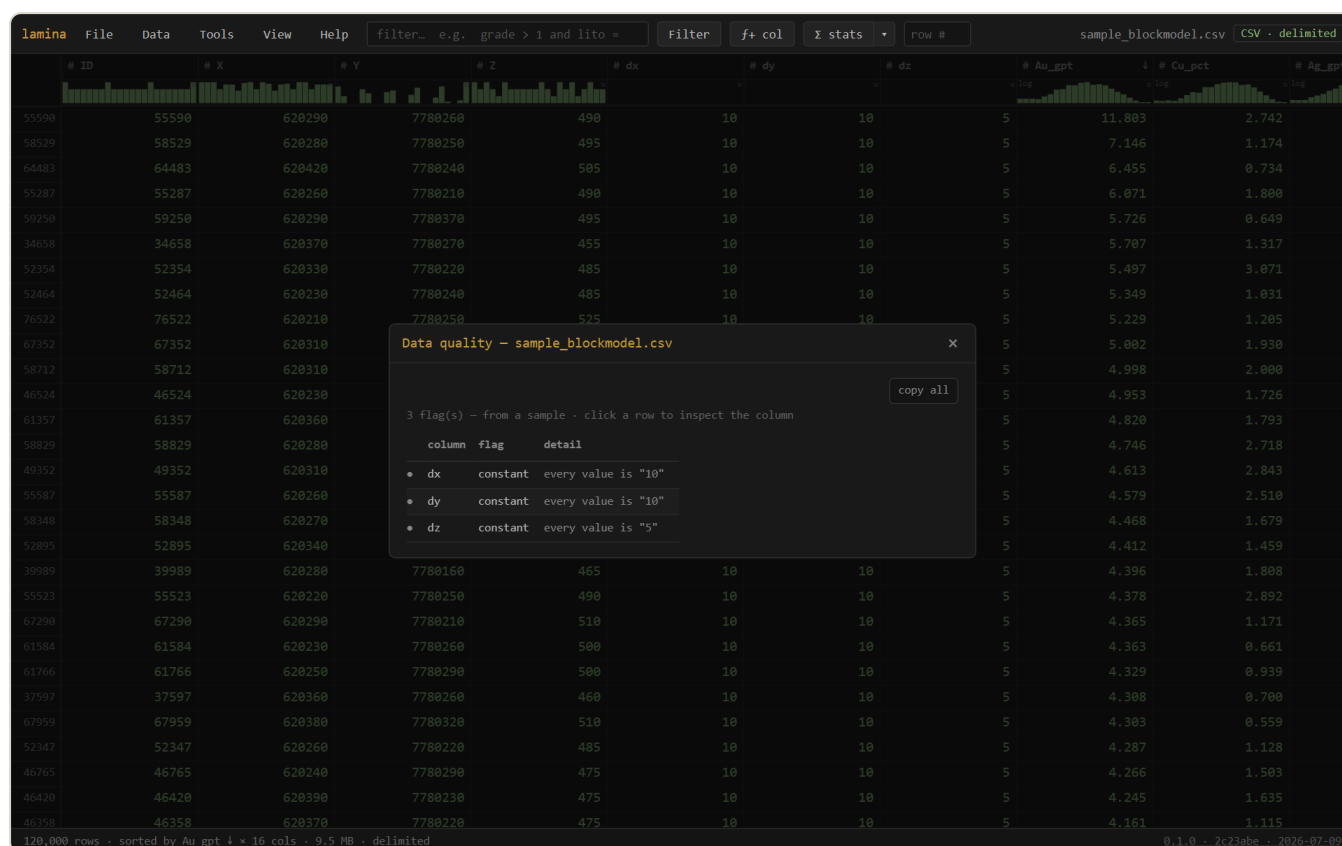
Grade-tonnage by domain with cutoff curves — tonnes falling, mean grade rising. Straight off the raw file, respecting the active filter.

10 Grid summary

Tools → **Grid summary...** infers a block model's **grid** from its centroid columns: origin · block size · count per axis, coordinate **ordering** (fastest/slowest), **sub-block detection**, parent-cell count and **fill %**. "No regular grid" is itself an answer — scattered data, or a broken export. Copy as TSV.

11 Data quality

Tools → **Data quality...** scans for the quiet bugs that bite an estimate: **leading zeros lost** (a code like `007` read as a number — with a one-click *fix: treat as text*), **non-numeric values** hiding in numeric columns, **missing-value sentinels** (`-999` ...), thousands separators, whitespace padding, all-blank or constant columns, dates-as-text. Click a flag for that column's Statistics; ✓ when clean.



Data quality — the pre-flight check. Every flag is a thing that has, somewhere, quietly wrecked an estimate.

12 Lenses

Data → **Save lens...** saves your *view* — filter · sort · calculated columns · number formats · color scales · hidden columns · the grade-tonnage setup — as a small `.lamina` file. Not the data: the recipe for looking at it. **Apply lens...** (or just open a `.lamina`) re-applies it to the current file, matching columns **by name**; anything that doesn't match is skipped and said. One setup, reused across every export with the same schema — and shareable with a colleague.

13 Export

Data → **Export...** writes the current view — filtered, sorted, calculated columns included, chosen columns only — to CSV/TSV, **streamed** straight to the file you pick: never uploaded, never fully held in memory. Opening a `.dm` and exporting is the `.dm` → CSV converter, at any size.

14 Security: Sealed

lamina is **Sealed**: it makes **no network requests** — a browser security policy (`connect-src 'none'`) blocks every connection, so there is no telemetry, no cloud, no auto-update. It is also **WASM-free** with the runtime in the clear: the code is readable and auditable rather than opaque binary. Your data — models, assays, exports — stays on your device, and its security artifacts (capability declaration, SHA-256, SBOM) are build-enforced, not asserted.

Verify the build and read how to bring it through approval at gentropic.org/security.

15 Install & offline

lamina runs at gentropic.org/lamina — install it as an app from the browser (it's also on the **Microsoft Store** as *GCU Lamina*), or download `lamina.html` : one file, no installer, that you can keep, e-mail, drop on a shared drive, or run fully offline by double-clicking. Nothing to set up, nothing that phones home. Works in current Chrome / Edge / Firefox.

A Reference

Keyboard & mouse

Input	Action
<code>Ctrl+O</code>	open a file
<code>Ctrl+F</code>	find in the table — <code>Enter</code> / <code>Shift+Enter</code> next/prev, case · whole-cell · regex · column scope
<code>Enter</code> / <code>Esc</code> in the filter box	apply / clear the filter
click a header	sort (ascending → descending → off)
right-click a header	statistics · sort · filter by · number format · color scale · text/number · hide
right-click a cell	copy (with header / row #) · filter by this value · statistics
glyph: hover / click / drag / double-click	read · filter · brush a range · statistics
drag a column border / double-click it	resize / autofit
kind badge (top right)	change how the file is read
row # box	jump to a row
<code>Ctrl+C</code>	copy the selection as TSV

The filter language

Comparisons `=` `!=` `>` `>=` `<` `<=`, boolean `and` `or` `not` (C-style `==` `&&` also work), `between a and b`, `in (...)`, `contains` / `like` / `matches`, `is blank` / `is filled`, functions `round` `int` `abs` `sqrt` `log` `exp` `pow` `min` `max` `clamp` `bin` `if` `coalesce` `ifnum` `isnum` `isnan`. Bare words are columns, quoted words are text; bracket awkward names: `["Cu (ppm)"] > 30`. The same expressions define calculated columns and the grade-tonnage factors.

lamina — part of the Auditable project, Geoscientific Chaos Union. Engine: `@gcu/lamina` ; grid: `@gcu/loom` ; expressions: `@gcu/expr` ; archives: `fflate` · `fzstd` · `seek-bzip` (MIT). MIT.

Screenshots are generated from the app by an automated harness, so they track the current build. See the in-app **Help** menu for the terse quick reference.